

VINAY BHALERAO

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PROFESSIONAL SUMMARY

Aspiring AI Developer with hands on experience building RAG systems, fine-tuning Llama-3.2-3B using QLoRA, and developing AI-powered applications using Python, LangChain, Hugging Face, and vector databases. Delivered 4+ production workflow modules for Goa citizen services portal, serving thousands of users, using React.js, Node.js and PostgreSQL. Passionate about applying strong software engineering and Generative AI expertise to build scalable, production ready application.

TECHNICAL SKILLS

- Frontend : React.js, JavaScript (ES6+), HTML5, CSS3, Responsive Web Design
- Backend : Node.js, Express.js, REST APIs
- Databases : PostgreSQL, MySQL, MongoDB
- Tools & Platforms : Git, GitHub, GitLab , Postman, VS Code, SourceTree , SonarQube , Jaspersoft , Cursor
- AI / LLM Engineering : Python , OpenAI API, Ollama, LiteLLM, RAG, Prompt Engineering, Fine-tuning (QLoRA/PEFT), ChromaDB, Pydantic , FastAPI.

WORK EXPERIENCE

Trainee Software Engineer - Goa Electronics Limited

October 2025 – Present

Goa Online Portal Revamp — Government Digital Services Platform

- Delivered 4+ production-ready workflow modules (IPB, DG Set services) used daily by citizens and departmental officers across Goa, contributing to the modernisation of **legacy government portals**.
- Engineered dynamic multi-step forms with **complex validation logic**, status tracking, and approval workflow screens in React.js — reducing applicant submission errors by improving real-time field feedback.
- Integrated 10+ REST API endpoints with PostgreSQL-backed services, enabling seamless data retrieval, verification, and processing across applicant and **department-facing modules**.
- Collaborated across a cross-functional team of **engineers, QA, project managers**, and **government stakeholders** through full **development lifecycle** from requirement analysis and **development to UAT** and **production deployment**.
- Contributed to **SonarQube-reviewed codebases**, improving **code quality** and **maintainability** as part of the team's quality assurance process.

Key Achievements

Shipped multiple modules used in live **government application** processing within first 6 months. Improved portal usability by redesigning validation flows and user feedback mechanisms. Modernised 2 legacy portal services using React.js and PostgreSQL-backed APIs.

PROJECTS

Goa Online Portal Revamp – IPB Service

[React.js](#) · [PostgreSQL](#)

- Built applicant and departmental workflow modules end-to-end: form validations, **multi-stage approval flows**, **status tracking**, and database- integrated data screens and wrote PostgreSQL **stored procedures** (SPs). Contributed to testing, **debugging**, and production deployment — module is live and actively used in government application processing.

Price Prediction Benchmark - Fine-Tuned LLMs

[Python](#), [PyTorch](#), [scikit-learn](#), [XGBoost](#), [Hugging Face \(Datasets/Hub\)](#), [LiteLLM](#), [OpenAI API](#), [QLoRA \(PEFT\)](#), [Plotly](#)

- Built an end-to-end **Python** pipeline predicting product prices from text, using **Hugging Face** Datasets (20K+ Amazon listings), an **LLM-based** data cleaning stage (**LiteLLM**), a custom **PyTorch DNN**, and classical **ML** models (**Random Forest**, **XGBoost**). Fine-tuned **Llama-3.2-3B** via **QLoRA**, achieving 39.85 **MAE**

InsureLLM — Enterprise RAG Chatbot

[Python](#), [LiteLLM](#), [OpenAI API](#), [Ollama \(Llama 3.2\)](#), [ChromaDB](#), [Gradio](#), [Pydantic](#), [Tenacity](#)

- Built a **RAG chatbot** using Python, LiteLLM, **ChromaDB**, and **Gradio** with LLM-powered **semantic chunking**, dual-query retrieval with automatic query rewriting, and LLM-based reranking for context-aware Q&A over enterprise insurance documents. Evaluated retrieval and answer quality using **MRR**, **nDCG**, and an LLM-as-judge pipeline scoring accuracy, completeness, and relevance.

EDUCATION

Bachelor of Engineering in Information Technology

Goa University · CGPA : 8.3 - Honours in Data Science

2021 – 2025

CERTIFICATIONS

- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents - Udemy
- React – The Complete Guide 2025 - Udemy
- PostgreSQL Bootcamp: Go From Beginner to Advanced - Udemy